

Graph Prompt Clustering

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I. INTRODUCTION

Abstract—Due to the wide existence of unlabeled graph-structured data (e.g., molecular structures), the graph-level clustering has recently attracted increasing attention, whose goal is to divide the input graphs into several disjoint groups. However, the existing methods habitually focus on learning the graphs embeddings with different graph regularizations, and seldom refer to the obvious differences in data distributions of distinct graph-level datasets. How to characteristically consider multiple graph-level datasets in a general well-designed model without prior knowledge is still challenging. In view of this, we propose a novel Graph Prompt Clustering (GPC) method. Within this model, there are two main modules, i.e., graph model pretraining as well as prompt and finetuning. In the graph model pretraining module, the graph model is pretrained by a selected source graph-level dataset with mutual information maximization and self-supervised clustering regularization. In the prompt and finetuning module, the network parameters of the pretrained graph model are frozen, and a groups of learnable prompt vectors assigned to each graph-level representation are trained for adapting different target graph-level datasets with various data distributions. Experimental results across six benchmark datasets demonstrate the impressive generalization capability and effectiveness of GPC compared with the state-of-the-art methods.

Index Terms—Graph-level clustering, graph model pretraining, prompt and finetuning, mutual information maximization, self-supervised clustering.

Received 30 June 2024; revised 15 February 2025; accepted 13 March 2025. Date of publication 20 March 2025; date of current version 6 June 2025. This work was supported in part by the National Key Research and Development Program of China under Grant 2024YFA1011900, in part by the National Natural Science Foundation of China under Grant 62276277, in part by Open Project Program of Guangxi Key Laboratory of Digital Infrastructure under Grant GXDIOP2024011, and in part by Guangdong Basic and Applied Basic Research Foundation under Grant 2022B1515120059. Recommended for acceptance by J. Wang. (Corresponding authors: Pei-Yuan Lai; Chang-Dong Wang.)

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Digital Object Identifier 10.1109/TPAMI.2025.3553129

DATA clustering as a fundamental unsupervised learning technique in machine learning aims to organize a set of input data samples into several groups or clusters based on their similarities [1], [2], [3], where similar samples would get together into the same group while dissimilar ones pull away each other and belong to different groups. Data clustering techniques are widely used in many real-world applications [4], [5], such as community detection and domain adaptation. Traditional clustering methods mainly consist of k -means, spectral clustering, subspace clustering and their variants [6], [7], [8], where they merely focus on the common node-level clustering. In other words, they are limited to handling the input node-level data.

As a matter of fact, there widely exist graph-level/graph-structured data, e.g., molecular/protein structures and social networks. The graph-level data focus on multiple graphs as inputs, where each graph containing rich structural information is treated as a “sample”. More recently, some efforts have been made in exploring the underlying correlations between different graph-level data in a supervised or unsupervised way [9], [10], [11], [12], [13], [14], [15]. In this paper, we mainly concentrate on the challenging graph-level clustering task, where there are no ground-truth supervision signals. Take protein clustering as an example. Similar proteins would be together in the same category, and distinct categories of proteins are beneficial to the downstream analysis task, e.g., functional annotation.

The recent graph-level clustering methods generally attempt to learn the informative graph embeddings by the powerful graph neural networks with different graph regularizations. For instance, given multiple graphs as inputs, the work in [16] developed a graph-level contrastive clustering method, where the graph instances were studied by multi-granularity node information, and then two-level contrastive learning strategies (i.e., instance-level and cluster-level contrastive learning) were designed to regularize the graph embeddings. To avoid the cluster collapse, the cluster-level uniformity of different features on hypersphere was studied, and the robustness of generated pseudo labels was enhanced [17]. Cai et al. proposed a deep graph-level clustering method to facilitate the information balance between graph representations and its substructures, and provide reliable pseudo labels to ensure the discriminability of graph representations [18]. Compared to the common node-level clustering with additional neighborhood propagation signals, the emerging graph-level clustering is more challenging, since the individual graph instances are isolated from each other, and there are no explicit information propagation cues between different

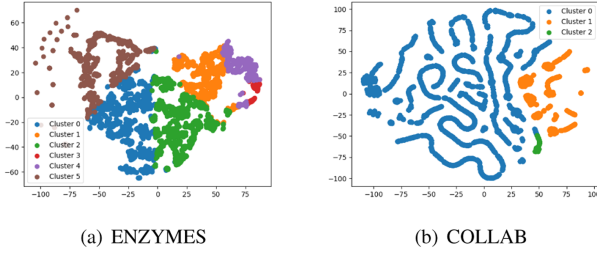


Fig. 1. T-SNE visualization of the data distributions from two graph-level datasets, i.e., ENZYMES and COLLAB. Herein, different dots stand for distinct graph instances, and different clusters are distinguishingly colored. The illustration results obviously demonstrate that data distributions of multiple graph-level datasets can be quite diverse.

graphs. More attempts about graph-level clustering to learn discriminative graph representations and make clear partitions are eager to be explored.

In general, the existing graph-level clustering methods habitually pay attention to learning the graph embeddings with different graph regularization, while seldom referring to the obvious differences in data distributions of distinct graph-level datasets. Empirically, the visualizations of two graph-level datasets are illustrated in Fig. 1, whose results further confirm the fact about differences in data distributions from distinct graph-level datasets. In other words, there are various types of data structures and cluster shapes. Hence, a natural question arises: *without prior knowledge, how can we characteristically consider multiple graph-level datasets in a general well-designed model.*

To address this question, in this paper, we devise a Graph Prompt Clustering (GPC) method, where a “pretraining-prompt-finetuning” scheme is adopted. In particular, the prompt learning is used for adapting different graph-level datasets with various data distributions in the generalized GPC model. To be specific, GPC consists of two main modules, i.e., graph model pretraining as well as prompt and finetuning. In the graph model pretraining module, the graph model is pretrained by a selected source graph-level dataset with mutual information maximization and self-supervised clustering regularization. In the prompt and finetuning module, the network parameters of the pretrained graph model are frozen, and a groups of learnable prompt vectors assigned to each graph-level representation are trained for adapting different target graph-level datasets with various data distributions. After that, we can directly obtain the final cluster assignment according to the optimal soft distribution.

The main contributions of this work are outlined as follows:

- To the best of our knowledge, it is the first attempt to introduce a “pretraining-prompt-finetuning” scheme in the challenging graph-level clustering task.
- We propose a novel graph-level clustering method, GPC, to empirically consider the differences in data distributions of distinct graph-level datasets, thus naturally leveraging the emerging prompt learning.
- Experimental results and in-depth analysis indicate the impressive generalization capability and effectiveness of GPC across six benchmark datasets compared with the state-of-the-art graph methods.

The rest of this work is organized as follows. Section II reviews some related work. In Section III, we provide the basic notations and preliminaries of graph-level clustering. The idea and overall framework of Graph Prompt Clustering model is elaborated in Section IV. In Section V, the extensive experimental results and in-depth analysis on several benchmark datasets are shown. Finally, we conclude this paper in Section VI.

II. RELATED WORK

Graph clustering algorithms: In general, there would be two different types of data scenarios considered in the graph clustering style algorithms, i.e., node-level data and graph-level data. On the one hand, the goal of node-level clustering is to recover the underlying correlations between nodes in the graph structure and partitioning the nodes into different categories. To combine the information from both node attributes and graph structure, the powerful graph neural networks are often adopted to learn a low-dimensional embedding by a neighborhood aggregation strategy. For instance, the work in [19] developed a structural deep clustering network model, in which the structural information was integrated for better clustering. In [20], a dynamic information fusion module was further designed to handle the attribute and graph structure information. Liu et al. proposed to learn the discriminative representation by a dual correlation reduction module in a siamese network-based framework [21]. Additionally, there are also some graph auto-encoder/variational graph auto-encoder based clustering models. The work in [22] attempted to capture the confidences of the neighboring nodes, and learn a compact representation by encoding the graph structure and node attributes based on the graph auto-encoder architecture. In [23], an adversarially regularized model was designed to learn graph embedding. In particular, more detailed information can refer to [24], [25], where a comprehensive perspective of deep (graph) clustering is reviewed. On the other hand, given multiple graphs as inputs, the graph-level clustering aims to separate them into several disjoint groups, where similar graphs are in the same group and dissimilar graphs belong to distinct groups. Recently, several associated attempts are made to capture the discriminative graph representations with multiple graph regularizations [16], [17], [18]. Nevertheless, the obvious differences in data distributions of distinct graph-level datasets are hardly referred and considered in the existing studies to improve the generalization of graph-level clustering.

Pretrained GNN models: Impressive performances achieved by the pre-trained models in natural language processing and computer vision motivate the recent development of the pre-trained GNN models [26]. To be specific, based on the given pretraining graphs, the self-supervised strategies are generally used to obtain meaningful representations during the network training process. For instance, the work in [27] trained the graph representations by choosing the edge prediction as the target task. Sun et al. aimed to capture the informative node or graph representations, in which the mutual information between structure-level and substructure-level representations were maximized [14]. Another widely used pretraining scheme for GNN methods is graph contrastive learning. Accordingly, various

TABLE I
DESCRIPTORS OF NOTATIONS USED THROUGHOUT THIS PAPER

$S = \{G_1, G_2, \dots, G_n\}$	Source graph-level dataset
$\mathcal{T} = \{G_1, G_2, \dots, G_{n'}\}$	Target graph-level dataset
$G_i = (\mathcal{V}_i, \mathcal{E}_i, X_i)$	The i -th graph instance
$\mathcal{V}_i$	Node set of G_i
$\mathcal{E}_i$	Edge set of G_i
X_i	Node feature matrix of G_i
$h_v^{(k)}$	Representation of node v at the k -th layer
h_φ^v	Node-level representation of node v
$H_\varphi(G_i)$	Graph-level representation of graph G_i
n_{G_i}	The number of graph nodes in G_i
φ	Parameter set of GNN
$I_{\varphi, \phi}$	Mutual information estimator
ϕ	Parameters of mutual information estimator
z_i	Latent representation of the graph G_i
$f_\psi(\cdot)$	Clustering projector parameterized by ψ
Q	Soft distribution
D	Target distribution
p_i	The i -th learnable prompt vector
$H_\varphi^*(G_i)$	Prompted graph-level representation of G_i
$\{p_1^b, p_2^b, \dots, p_k^b\}$	Basis vectors set
k	The number of basis vectors
a_j	The j -th learnable linear projection
$\alpha_{i,j}$	Weight of j -th basis vector to G_i

studies about different augmentation strategies for contrastive learning are made [28], [29]. Diverse pretraining strategies for GNN methods are especially designed by the characteristic objectives [30].

Prompt-based tuning methods: Originated from natural language processing, the prompt-based tuning models aiming to encourage the adaptation of pretrained language models to multiple objective tasks are widely studied [31]. Especially, the remarkable performances are achieved by the soft prompt based researches, where the parameters of the pretrained models are frozen, and additional learnable variants are introduced in the input/representation space [32], [33]. With such a design, the compatibility between newly inputs and pretrained models can be well enhanced. Motivated by the success of prompt-based tuning models in the language field, some related efforts are made in GNN. The earlier works pretrained the models by edge prediction, based on which the prompt-based tuning methods were applied [34], [35]. Furthermore, a universal graph prompt feature method was developed for the pretrained GNN models with any pretraining scheme [36], where the prompted graph was constructed in an adaptive manner.

III. NOTATIONS AND PRELIMINARIES

In this section, some notations and preliminaries about graph clustering are introduced. In particular, we summarize the notations used throughout this paper in Table I.

Definition 1 (Topological graph data): Given a graph $G = (\mathcal{V}, \mathcal{E}, X)$, we use $\mathcal{V}$ and $\mathcal{E}$ to denote the set of nodes and edges, respectively. $X \in \mathbb{R}^{m \times d}$ is the node feature matrix associated

with each node $v \in \mathcal{V}$, where m is the number of nodes, and d denotes the feature dimension.

Definition 2 (Graph-level data): The graph-level data is denoted as $\mathcal{G} = \{G_1, G_2, \dots, G_n\}$, where $G_i = (\mathcal{V}_i, \mathcal{E}_i, X_i)$ and n denotes the number of graphs.

Definition 3 (Graph-level data clustering): The graph-level data clustering attempts to separate multiple graphs from the input $\mathcal{G}$ into several disjoint categories, in order that semantically similar graphs are grouped together (into the same category) in an unsupervised setting and vice versa.

Definition 4 (Graph neural network): Given a graph $G = (\mathcal{V}, \mathcal{E}, X)$, graph neural network (GNN) can be usually adopted to learn a discriminative graph-level representation, where the node feature information and structural information are jointly considered. Specifically, the local node neighboring information is aggregated by message-passing mechanism, and the feature representation of node v at the k th layer is written as,

$$h_v^{(k)} = \text{Update}^{(k)} \left(h_v^{(k-1)}, \text{Aggregate} \left(h_u^{(k-1)}_{u \in \mathcal{N}(v)} \right) \right), \quad (1)$$

where $\mathcal{N}(v)$ denotes the neighbors of node v . Note that the initial feature representation (i.e., $h_v^{(0)}$) is the original node feature (i.e., x_v). Especially, after K layers of iteration, a set of feature representations of node v at all depths (i.e., $\{h_v^{(k)}\}_{k=1}^K$) are obtained. To consider different feature information at distinct scales oriented to each node, the corresponding representations are then summarized [37], i.e.,

$$h_\varphi^v = \text{Concat} \left(\left\{ h_v^{(k)} \right\}_{k=1}^K \right), \quad (2)$$

where h_φ^v denotes the final node-level representation of node v , and φ is the parameter set of a K -layers GNN. The graph-level representation can be then computed by combining all node-level representations, i.e.,

$$H_\varphi(G) = \text{Readout} \left(\left\{ h_\varphi^v \right\}_{v=1}^m \right), \quad (3)$$

where m is the number of nodes in G , and the Readout function is often achieved by sum, average or adaptive pooling function. Note that in this paper, the sum readout strategy is employed.

IV. APPROACH

A. Overview

Due to the differences in data distributions of distinct graph-level datasets, the basic idea of GPC is to develop a generalized graph-level clustering method appropriate to multiple graph-level datasets by a well-designed “pretraining-prompt-finetuning” scheme. For clarity, the overall framework is illustrated in Fig. 2.

Within this model, there are two main modules: graph model pretraining as well as prompt and finetuning. With a source graph-level dataset as an input, the graph model pretraining module adopts a graph isomorphism network (i.e., GNN herein) to learn the graph-level representations, which are first computed by combining the graph-oriented node-level representations. The mutual information between the graph-level representations and the graph-oriented node-level representations are then

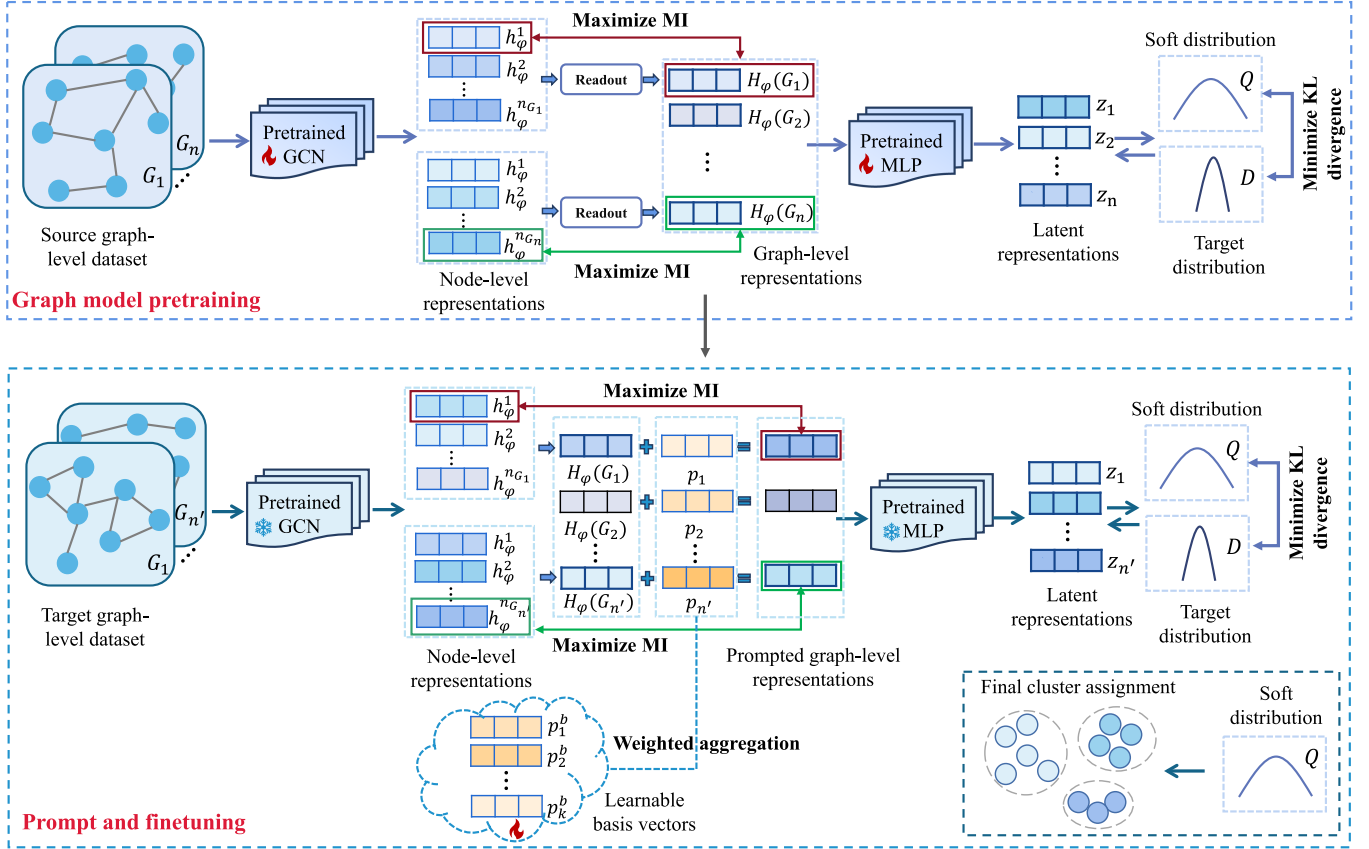


Fig. 2. Illustration of the proposed GPC method. There are two main modules, i.e., graph model pretraining as well as prompt and finetuning. In the graph model pretraining module, the graph model is pretrained by a source graph-level dataset with mutual information maximization (i.e., Maximize MI) and self-supervised clustering regularization (i.e., Minimize KL divergence). In the prompt and finetuning module, the network parameters of the pretrained graph model are frozen, and a groups of learnable prompt vectors assigned to each graph-level representation are trained for adapting different target graph-level datasets with various data distributions. The final cluster assignment can be finally obtained based on the optimal soft distribution.

maximized to enhance the representation learning. To match the downstream clustering task, the latent representations can be obtained based on the graph-level representations by a MLP (Multi-layer Perceptron) based clustering projector. And the soft labels generated by latent representations are used to self-supervise the cluster-oriented discriminative representations. When there newly comes the target graph-level dataset, the prompt and finetuning module is designed, where the network parameters of the pretrained model are frozen, and an independent learnable prompt vector is assigned to each graph-level representation. Note that the regularizations of this module are the same as the ones of graph model pretraining module, and is directly used to guide the representation learning as well as prompt learning. Finally, the final graph-level clustering assignment can be obtained based on the optimal soft distribution.

B. Graph Model Pretraining

Graph-level representation learning: Given a source graph-level dataset $\mathcal{S} = \{G_1, G_2, \dots, G_n\}$ with empirical probability distribution $\mathbb{P}$ on the data space, we can certainly obtain the node-level representation h_φ^v of node $v \in \mathcal{V}_i$ and the graph-level representation $H_\varphi(G_i)$ for each G_i by GNN according

to Definition 4, where φ denotes the parameter set of the used K -layers GNN. In order to boost the representation learning in an unsupervised manner, the mutual information between the graph-level representation and the graph-oriented node-level representations are maximized [14]. To be specific, the mutual information estimator $I_{\varphi, \phi}$ on the graph-level and node-level pairs can be defined as,

$$\mathcal{L}_{R|\varphi, \phi} \triangleq \arg \max_{\varphi, \phi} \sum_{G_i \in \mathcal{S}} \frac{1}{n_{G_i}} \sum_{v \in \mathcal{V}_i} I_{\varphi, \phi}(h_\varphi^v; H_\varphi(G_i)), \quad (4)$$

where n_{G_i} records the number of graph nodes in G_i . ϕ are the parameters of a discriminator network T_ϕ used to parameterize $I_{\varphi, \phi}$. More specifically, the Jensen-Shannon mutual information estimator [38] is adopted to formulate $I_{\varphi, \phi}$, i.e.,

$$I_{\varphi, \phi}(h_\varphi^v(G_i); H_\varphi(G_i)) := \mathbb{E}_{\mathbb{P}} \left[-\text{sp}(-T_{\varphi, \phi}(h_\varphi^v(s); H_\varphi(s))) \right] \\ - \mathbb{E}_{\mathbb{P} \times \tilde{\mathbb{P}}} [\text{sp}(T_{\varphi, \phi}(h_\varphi^v(s'); H_\varphi(s)))], \quad (5)$$

where s is the input positive sample, and s' is the negative one sampled from a distribution $\tilde{\mathbb{P}}$ identical to the empirical probability distribution $\mathbb{P}$ on the input data space. $\text{sp}(a) =$

$\log(1 + e^a)$ denotes the softplus function. Note that the negative samples are generated by using combinations of graph-level and node-level representations across all graph samples in a batch. To be exact, after maximizing the mutual information between the graph-level and graph oriented node-level representations, the graph-level representations can be facilitated to contain more semantics information shared between the graph oriented node-level representations.

Self-training graph-level clustering: To capture the latent cluster-oriented representations suitable to the downstream clustering task, a MLP based clustering projector is connected with the graph-level representations. Formally, the latent representations can be obtained by projecting the graph-level representations to the low-dimensional latent space, whose formulation is written as,

$$z_i = f_\psi(H_\varphi(G_i)), \quad (6)$$

where z_i stands for the latent representation of the graph instance G_i . $f_\psi(\cdot)$ denotes the MLP based clustering projector parameterized by the parameter ψ . To guide the learning of latent representations, a self-training clustering objective is suggested to be formulated, in which the highly confident samples are treated as soft labels to guide the clustering process. Especially, based on the latent representations, the self-training clustering objective is defined as,

$$\mathcal{L}_{C|\varphi,\psi} = KL(D||Q) = \sum_i \sum_j d_{ij} \log \frac{d_{ij}}{q_{ij|\varphi,\psi}}, \quad (7)$$

where $KL(\cdot||\cdot)$ represents the Kullback-Leibler divergence between two distributions Q and D . Q denotes the soft distribution recording the clustering assignment probability about each graph instance belonging to different clusters, and similarly D stands for the target distribution. To be specific, q_{ij} can be calculated by the Student's t-distribution, whose formulation is written as,

$$q_{ij|\varphi,\psi} = \frac{(1 + \|z_i - \mu_j\|_2^2)^{-1}}{\sum_{j'} (1 + \|z_i - \mu_{j'}\|_2^2)^{-1}}, \quad (8)$$

where μ_j is the j th cluster centroid computed by k -means algorithm on $Z = \{z_1, z_2, \dots, z_n\}$. Based on Q , the target distribution is then formulated as,

$$d_{ij} = \frac{q_{ij|\varphi,\psi}^2 / \sum_i q_{ij|\varphi,\psi}}{\sum_{j'} q_{ij'|\varphi,\psi}^2 / \sum_i q_{ij'|\varphi,\psi}}. \quad (9)$$

By raising the soft distribution Q to the second power, the target distribution D is encouraged to show a more clear gap between high and low probability values in Q , which is then used as the pseudo labels for the guidance of optimizing Q . With such a self-training clustering objective minimization, Q will get close to D , where the high confident assignment can be emphasized.

Objective function design: To jointly optimize the graph-level representation learning and self-training graph-level clustering, we define the objective function as,

$$\mathcal{L} = -\mathcal{L}_{R|\varphi,\phi} + \mathcal{L}_{C|\varphi,\psi}. \quad (10)$$

Algorithm 1: Graph Prompt Clustering.

Input: Source graph-level dataset $\mathcal{S} = \{G_1, G_2, \dots, G_n\}$, target graph-level dataset $\mathcal{T} = \{G_1, G_2, \dots, G_{n'}\}$, basis vector number k , epoch n_{iter} .

Graph model pretraining:

- 1: Randomly initialize the parameters φ, ψ in this module.
- 2: **for** $\forall i = 1, \dots, n_{\text{iter}}$ **do**
- 3: Optimize each node-level representation h_φ^v of different graph instances from $\mathcal{S}$ according to (2).
- 4: Optimize the graph-level representation $H_\varphi(G_i)$ of each graph instance from $\mathcal{S}$ according to (3).
- 5: Optimize each cluster-oriented latent representation z_i according to (6).
- 6: Compute the mutual information loss $\mathcal{L}_{R|\varphi,\phi}$ according to (4).
- 7: Compute the self-training clustering loss $\mathcal{L}_{C|\varphi,\psi}$ according to (7).
- 8: Compute the additive objective function loss $\mathcal{L}$ according to (10).
- 9: Perform the gradient descent to optimize $\mathcal{L}$.
- 10: **end for**
- 11: **Return** φ, ψ .

Prompt and finetuning:

- 1: Freeze φ, ψ .
- 2: Randomly initialize the basis vectors $\{p_1^b, p_2^b, \dots, p_k^b\}$ and linear projections $\{a_1, a_2, \dots, a_k\}$.
- 3: **for** $\forall i = 1, \dots, n_{\text{iter}}$ **do**
- 4: Optimize each node-level representation h_φ^v of different graph instances from $\mathcal{T}$ according to (2).
- 5: Optimize the graph-level representation $H_\varphi(G_i)$ of each graph instance from $\mathcal{T}$ according to (3).
- 6: Optimize the prompted graph-level representation $H_\varphi^*(G_i)$ according to (11).
- 7: Optimize each cluster-oriented latent representation z_i based on $\{H_\varphi^*(G_i)\}_{i=1}^{n'}$ according to (6).
- 8: Compute the mutual information loss $\mathcal{L}_{R|\varphi,\phi}$ according to (4).
- 9: Compute the self-training clustering loss $\mathcal{L}_{C|\varphi,\psi}$ according to (7).
- 10: Compute the additive objective function loss $\mathcal{L}$ according to (10).
- 11: Perform the gradient descent to optimize $\mathcal{L}$.
- 12: **end for**
- 13: Compute the prompt vectors $\{p_1, p_2, \dots, p_{n'}\}$ according to (12).
- 14: **Return** $\{p_1, p_2, \dots, p_{n'}\}$.

Output: Obtain the final cluster assignment of $\mathcal{T}$ based on the optimal soft distribution.

C. Prompt and Finetuning

Prompt learning and finetuning: Based on the graph model pretraining module, when there newly comes the target graph-level dataset $\mathcal{T} = \{G_1, G_2, \dots, G_{n'}\}$ probably with different data distributions, a prompt and finetuning module is seamlessly designed, in which the network parameters of the pretrained model are frozen, and independent learnable prompt vector are constructed and respectively assigned to each learned graph-level representations. More precisely, the node-level

representations and initial graph-level representations of the input target graph-level dataset are first learned by the frozen GCN. And then each prompted graph-level representation of graph instance G_i , denoted as $H_\varphi^*(G_i)$, is constructed and can be generated by adding the learnable prompt vector p_i to the initial graph-level representation $H_\varphi(G_i)$, i.e.,

$$H_\varphi^*(G_i) = H_\varphi(G_i) + p_i. \quad (11)$$

In order to avoid learning a series of prompt vectors $\{p_1, p_2, \dots, p_n\}$, we train only a set of k independent basis vectors, i.e., $\{p_1^b, p_2^b, \dots, p_k^b\}$, where k is a predefined value or can also be adjusted according to the input graph-level datasets. Particularly, each prompt vector p_i can be then calculated by employing the weighted aggregation of these basis vectors, which is formulated as,

$$p_i = \sum_j^k \alpha_{i,j} p_j^b, \quad \alpha_{i,j} = \frac{\exp(a_j^T H_\varphi(G_i))}{\sum_l^k \exp(a_l^T H_\varphi(G_i))}, \quad (12)$$

where a_j denotes the j th learnable linear projection. After that, we can use the constructed prompt vectors to generate each prompted graph-level representations as described in (11).

It is worth mentioning that the regularizations of this module are the same as the ones of graph model pretraining module, and can be directly used to guide the representation learning as well as prompt learning. In other words, the mutual information maximization between the prompted graph-level representation and the graph-oriented node-level representations is calculated. Based on the prompted graph-level representations, the latent cluster-oriented representations would be captured by the frozen MLP based clustering projector. And the self-training clustering objective is similarly defined to self-supervise the cluster-oriented discriminative representations.

After completing the optimization of the representations and prompts, we can obtain the final cluster assignment of target graph-level dataset based on the optimal soft distribution.

Remarks: 1) In this work, the prompt vectors are the specific characteristic cues learned from the target graph-level data, which act as adaptive interfaces between the frozen pre-trained graph model and the target datasets with diverse data distributions. The prompts are designed to “steer” the graph representations toward capturing dataset-specific characteristics (e.g., variations in node/edge attributes or structural patterns) by aligning the semantic space of the source and target datasets. Furthermore, the prompts in GPC can directly modulate the latent representations to bridge the domain gap between source and target datasets, while preserving the pretrained knowledge in the frozen backbone. 2) Different from the prompt-based methods, to avoid the interference caused by task discrepancies and focus on evaluating the adaptation capability of the proposed GPC method towards distinct graph-level datasets, the training tasks between the graph model pretraining module and the prompt and finetuning module are the same. Such a design ensures that the model can adapt to different data distributions through

prompt learning and finetuning in an unsupervised setting while preserving the pretrained knowledge, without modifying the pretrained parameters.

D. Algorithm Summary

To sum up, the optimization process of GPC is summarized in Algorithm 1. In the proposed GPC model, considering the differences in data distributions of distinct graph-level datasets, its main idea is to develop a generalized graph-level clustering method appropriate to multiple graph-level datasets by a well-designed “pretraining-prompt-finetuning” scheme. There are two main modules, i.e., the graph model pretraining for the source graph-level dataset as well as prompt and finetuning for the target graph-level dataset. Specifically, the regularizations of these two modules are the same, and we use the gradient descent strategy to optimize the additive objective function $\mathcal{L}$, where the mutual information loss $\mathcal{L}_R$ well boost the graph-level representation learning, and the self-training clustering loss $\mathcal{L}_C$ provides the self-supervision signals to guide the learning of cluster-oriented latent representations. Their differences lie in that for the prompt and finetuning module, the network parameters of the pretrained model are frozen, and independent learnable prompt vector are constructed and respectively assigned to each learned graph-level representations.

For algorithm reproducibility, the source code of GPC is publicly available at <https://github.com/ManshengChen/Code-for-GPC-master>.

V. EXPERIMENTS

In this section, to validate the effectiveness of the proposed GPC method, experiments are well-designed on several benchmark graph-level datasets compared with the state-of-the-art graph clustering methods.

A. Experimental Setup

Datasets: Seven real-world graph-level datasets are used in the experiments, especially consisting of one source graph-level dataset (i.e., BZR) and six target graph-level ones (i.e., PTC-MM, ENZYMES, COX2, NCI1, NCI109 and COLLAB). Detailed information about these benchmark datasets will be introduced in the following.

- *BZR* as the ligand dataset for benzodiazepine receptor consists of 405 graphs, and there is an average of 35.75 nodes and 38.36 edges for each graph instance. According to the activity and inactivity of compounds, these graphs are divided into two categories in the experiments.
- *PTC-MM* is extracted from the original PTC dataset. Especially, PTC-MM is composed of 336 compounds with average 13.97 nodes and 14.32 edges. Based on the carcinogenicity to rodents, these compounds are both divided into two categories.
- *ENZYMES* consists of 600 protein data with 100 protein data being in a group. Particularly, each protein is treated as a graph, and there is an average of 32.63 nodes and 62.14 edges for each graph instance.

TABLE II
DESCRIPTORS OF SIX BENCHMARK GRAPH-LEVEL DATASETS CONSISTING OF ONE SOURCE GRAPH-LEVEL DATASET AND FIVE TARGET GRAPH-LEVEL ONES

	Datasets	Types	#Graphs	#Nodes	#Edges	#Classes
Source	BZR	Molecules	405	35.75	38.36	2
Target	PTC-MM	Molecules	336	13.97	14.32	2
	ENZYMES	Molecules	600	32.63	62.14	6
	COX2	Molecules	467	41.22	43.45	2
	NCI1	Molecules	4,110	29.87	32.30	2
	NCI109	Molecules	4,127	14.29	14.69	2
	COLLAB	Social Networks	5,000	74.49	2457.78	3

- *COX2* contains 467 inhibitors with average 41.22 nodes and 43.45 edges. According to the activity and inactivity of compounds, they are directly grouped into two categories.
- *NCI1* consists of 4110 chemical compounds, and there is an average of 29.87 nodes and 32.30 edges. Especially, these compounds are divided into either positive or negative groups concerning about the cell lung cancer.
- *NCI109* contains 4127 chemical compounds with average 14.29 nodes and 14.69 edges. Similarly, these compounds are categorized into the positive or negative groups related to the ovarian cancer.
- *COLLAB* is a graph-level dataset about social networks, which is composed of 5000 graphs. For each graph, there is an average of 74.49 nodes and 2457.78 edges. And in this experiment, these graphs are grouped into three categories.

For clarity, the descriptions of these seven benchmark graph-level datasets are summarized in Table II. Specifically, the used graph-level datasets can be downloaded from <https://chrsmrrs.github.io/datasets/docs/datasets/>, and in our experiments, the used datasets are directly used with no additional processing.

Baselines: Four classic graph kernel methods and four unsupervised graph-level representation learning methods are introduced to validate the effectiveness of the proposed GPC method. More precisely, the introductions about these compared baselines are shown as follows.

- *Random Walk Kernel (RWK)* [10] attempts to perform random walks on pairwise graphs, and calculate the number of matching walks.
- *Weisfeiler-Lehman Kernel (WLK)* [39] is an efficient feature extraction strategy based on the Weisfeiler-Lehman test of graph isomorphism.
- *Shortest Path Kernel (SPK)* [40] is an expressive graph kernel based on the shortest path, whose computation is in polynomial time.
- *Lovasz-Theta Kernel (LTK)* [41] aims to capture global graph properties via the Lovasz number and the corresponding orthonormal representation.
- *InfoGraph* [14] attempts to learn informative graph-level representations by maximizing the mutual information between the graph-level representation and the substructure representations.
- *Graph Contrastive Learning (GraphCL)* [28] aims to capture the unsupervised graph representations based on a graph contrastive learning scheme, where different types of graph augmentations are specifically considered.

- *Joint Augmentation Optimization (JOAO)* [29] enable to adaptively and automatically choose data augmentations when performing graph contrastive learning on graphs.
- *Deep Graph-Level Clustering (DGLC)* [18] attempts to capture graph-level representations and make partitions on graphs in a unified framework.

Note that for the graph kernel methods (i.e., RWK, WLK, SPK and LTK), the spectral clustering (SC) is employed on the generated affinity matrices to achieve the final clustering performance. For the unsupervised graph-level representation learning methods (including InfoGraph, GraphCL, JOAO), the spectral clustering (SC) and *k*-means (KM) method are adopted on the learned graph representations to obtain the clustering results. And DGLC is an end-to-end graph-level clustering method, capable of directly obtaining the final performances.

Implementation: In the experiments, the source codes of the compared methods are downloaded, and their best performances are tuned by the suggested ranges of parameters. For fair comparison, the same network architecture as the compared unsupervised graph-level representation learning methods is adopted, where the graph isomorphism network is specially used as the backbone GNN. The MLP based clustering projector is a two-layer fully-connected network. During the training of the model, we use an Adam optimizer with the learning rate setting to 1e-4, and the batch size is 128. In addition, to ensure the generalization of GPC, on the source graph-level dataset (i.e., BZR), the layer number of GNN is set to 3, the hidden dimension of each GNN layer is 32, and the latent dimension of MLP projector is 25. Note that on the target graph-level datasets, the backbone GNN and MLP projector are frozen. In other words, their corresponding network architecture and parameters are fixed.

Evaluation metrics: Three commonly used evaluation metrics are adopted, i.e., accuracy (ACC), normalized mutual information (NMI) and adjusted rand index (ARI). For these metrics, they are exactly in the range of 0 to 1, and higher values indicate better performances. In our experiments, all experiments are run ten times, and the average performances presented in percentages would be reported.

B. Comparison Results

The quantitative results of the proposed GPC method against the state-of-the-art graph clustering methods are recorded in Tables III and IV, where the best performance is highlighted

TABLE III
EXPERIMENTAL RESULTS ON THE PTC-MM, ENZYMES AND COX2 DATASETS

Datasets	PTC-MM			ENZYMES			COX2		
Method	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI
RWK-SC	60.71	0.97	2.91	17.00	0.66	0.25	51.31	0.70	-0.92
WLK-SC	<u>62.20</u>	1.50	3.87	21.00	3.09	1.48	50.54	0.51	-0.40
SPK-SC	<u>62.20</u>	1.63	0.73	22.00	2.57	1.69	52.03	0.13	0.01
LTK-SC	61.19	0.73	1.09	17.00	0.42	0.00	<u>77.52</u>	0.26	0.17
InfoGraph-KM	61.48	2.35	3.61	22.06	2.40	1.25	56.74	3.30	0.17
InfoGraph-SC	61.96	2.12	<u>4.55</u>	23.75	4.64	<u>2.23</u>	70.37	3.56	1.92
GraphCL-KM	58.93	0.27	0.60	21.50	1.55	0.90	68.88	1.05	0.44
GraphCL-SC	62.09	2.14	3.36	<u>25.28</u>	4.75	2.03	75.01	1.24	2.39
JOAO-KM	59.04	0.21	0.98	21.66	1.60	0.94	70.56	1.19	0.44
JOAO-SC	62.41	2.00	4.28	24.65	<u>4.85</u>	2.07	76.46	1.43	2.35
DGLC	46.43	<u>2.42</u>	0.68	22.67	4.23	1.41	78.8	<u>3.87</u>	12.85
GPC	61.9	4.97	4.78	26.67	6.79	2.65	70.88	4.25	<u>3.42</u>

Note that the results are presented in percentages.

TABLE IV
EXPERIMENTAL RESULTS ON THE NCI1, NCI109 AND COLLAB DATASETS

Datasets	NCI1			NCI109			COLLAB		
Method	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI
RWK-SC	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
WLK-SC	50.05	0.00	0.00	50.39	0.01	0.00	53.20	1.96	0.53
SPK-SC	50.10	0.10	0.00	52.26	0.33	0.19	48.72	17.91	13.93
LTK-SC	N/A	N/A	N/A	50.47	0.01	-0.01	N/A	N/A	N/A
InfoGraph-KM	54.11	1.28	0.85	54.38	1.25	0.89	59.64	14.40	6.61
InfoGraph-SC	54.87	0.93	1.04	54.67	1.08	1.00	60.92	15.37	9.33
GraphCL-KM	55.37	0.47	0.99	55.37	1.79	2.11	58.02	17.81	11.33
GraphCL-SC	55.93	0.61	1.08	56.29	2.12	2.48	57.83	16.97	10.10
JOAO-KM	51.12	0.43	0.05	56.20	1.73	1.54	58.34	18.73	11.06
JOAO-SC	51.48	0.88	0.40	<u>56.30</u>	4.61	1.81	57.84	17.12	10.55
DGLC	<u>57.93</u>	5.13	<u>2.50</u>	54.71	1.92	1.55	59.34	<u>22.20</u>	11.26
GPC	59.66	<u>2.89</u>	3.71	57.43	<u>2.51</u>	<u>2.19</u>	<u>59.72</u>	23.06	<u>12.25</u>

Note that the results are presented in percentages.

in boldface while the second best performance is in underline. According to the experimental results, the following observations can be acquired.

- In general, the proposed GPC method achieves better performance on all the five benchmark target datasets compared with the other graph kernel methods and unsupervised graph-level representation learning methods. Especially on PTC-MM and ENZYMES, GPC outperforms the second best methods (i.e., DGLC and JOAO-SC) with 2.55% and 1.94% respectively in terms of NMI, which explicitly indicates the impressive generalization capability of GPC in graph-level clustering.
- The unsupervised graph-level representation learning methods relatively perform better than the graph kernel methods from an overall average perspective. It is worth noting that WLK-SC obtains satisfactory performance on PTC-MM while getting mediocre on the rest target datasets. Similarly, DGLC merely achieves impressive

clustering results on COX2 and NCI1. As a matter of fact, it is not easy for a method to acquire the remarkable performance on multiple graph-level datasets with probably different data distributions. However, the proposed GPC well do it by a seamless “pretraining-prompt-finetuning” scheme.

- Different post-processing techniques, i.e., k -means (KM) or spectral clustering (SC), have an obvious influence on the final results. On the whole, the method with SC often does better than itself with KM. Actually, no matter which technology we used, the discriminative cluster-oriented graph-level representations are the key points.
- Due to the inherent complexity and high variability of most molecular datasets, particularly challenging clustering is commonly made by these graph clustering methods, with low NMI and ARI values on these datasets. Compared to the relevant baselines, the proposed GPC method with a “pretraining-prompt-finetuning” scheme generally demonstrates its relative effectiveness on all the five benchmark

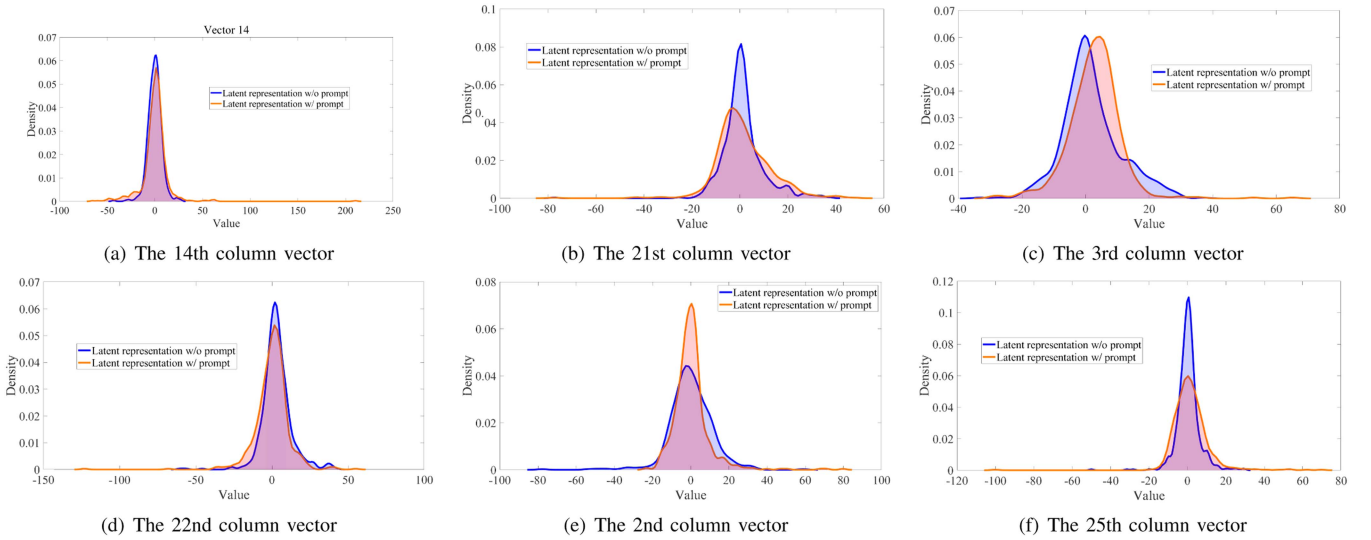


Fig. 3. Empirical study on the ENZYMES dataset. Six column vectors of the latent cluster-oriented representations without/with prompt (learning) are randomly selected to understand the the distribution characteristics of data via the kernel density estimation.

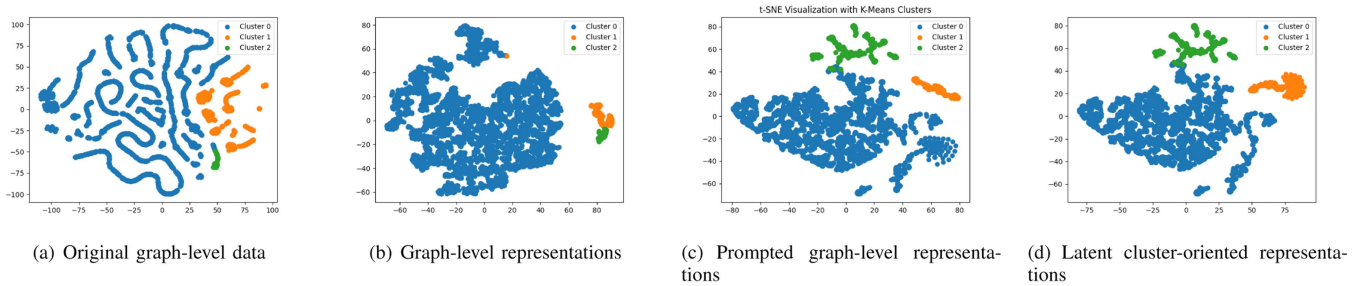


Fig. 4. T-SNE visualization analysis about the original graph-level data, graph-level representations, prompted graph-level representations and latent cluster-oriented representations on the COLLAB dataset.

target datasets, which further confirms the capability to consider distinct graph-level datasets in a general well-designed model without prior knowledge.

C. Empirical Study

In this subsection, an empirical study is conducted on the ENZYMES dataset to understand the exact distribution of the embedding data, by which the influences brought by the prompt learning can be obviously obtained. Note that to better demonstrate the results of the kernel density estimation, only the ENZYMES dataset with the largest number of classes is adopted in this experiment. Especially, we randomly select six column vectors on the same dimensions of the latent cluster-oriented representations without/with prompt learning, and then make kernel density estimation [42], [43] to better understand the distribution characteristics of data, whose results can be found in Fig. 3. According to these subfigures, we can find that the column vector on the same dimension selected from the latent representation with prompt learning exhibit a larger fluctuation range in activation values compared to the one selected from

the latent representation without prompt learning. For instance, for the 25th column vector (illustrated in Fig. 3(f)), the presented value of the latent representation with prompt learning ranges in $[-105, 75]$, while the one of the latent representation without prompt learning ranging in $[-52, 32]$. In other words, the corresponding variance is greater, and this also implies larger distances between distinct categories, explicitly resulting in better clustering performance with the prompt learning strategy.

D. Visualization Results

In order to better investigate the learned representations as well as the generalization capability of GPC, we take COLLAB as an example and make the corresponding visualization in Fig. 4 via t-SNE [44]. In this figure, the visualization about the original graph-level data, graph-level representations, prompted graph-level representations and latent cluster-oriented representations on COLLAB are conducted. Though the structure observed from the original graph-level data is labyrinthine, a clear structure of latent cluster-oriented representations can be well recovered by GPC based on the prompted graph-level representations, where

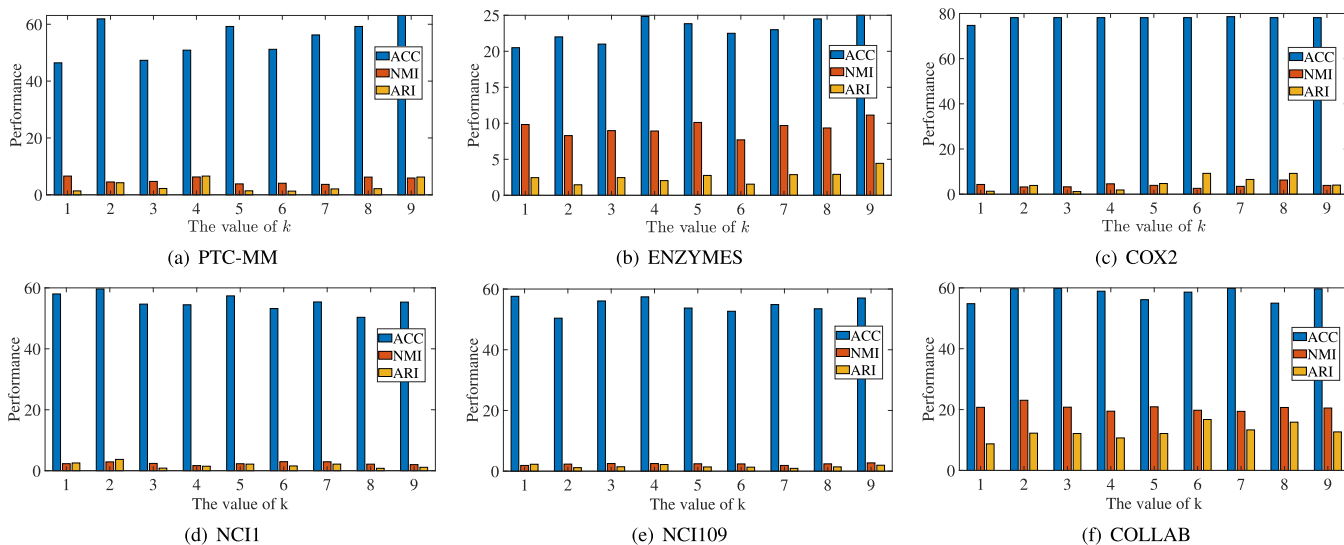


Fig. 5. Parameter sensitivity analysis on the six benchmark graph-level datasets.

exactly the prompts are designed for characterizing different graph data distributions in a generalized graph-level clustering model. Meanwhile, according to the visualizations about graph-level representations and the prompted graph-level representations, it can be observed that the boundary of different cluster structures become gradually obvious with the adoption of prompts. Note that in this visualization experiment, the reason why we only choose COLLAB is that this dataset is about social networks, quite different from the source molecules dataset (i.e., BZR), and its overall performances achieved by different baselines are more remarkable than those of the other datasets, and then the visualization results can be better presented.

E. Parameter Sensitivity

In the graph model pretraining module, the source graph-level dataset, i.e., BZR, is used to pretrain the network parameters by the well-designed loss functions. Subsequently, given the target graph-level datasets, the prompt and finetuning module aims to directly achieve a remarkable graph-level clustering by the frozen pretrained model and learnable prompt vectors. Therefore, in this subsection, only the number of basis vectors k related to the prompt is studied, whose experimental results in terms of three metrics are illustrated in Fig. 5. According to this figure, we can observe that the proposed GPC method generally performs stably in the provided ranges of k . Meanwhile, it can be seen that the best performances on the six benchmark datasets can be achieved by the certain value of prompt. For instance, on PTC-MM, when the number of basis vectors k is 9, the best results are obtained.

VI. CONCLUSION

In this paper, considering the obvious differences in data distributions of distinct graph-level datasets, we developed a generalized method called Graph Prompt Clustering (GPC), where a “pretraining-prompt-finetuning” scheme is well-designed to

generally adapt to different graph-level datasets. In this model, the graph model is pretrained by a source graph-level dataset with mutual information maximization and self-supervised clustering regularization. For the prompt and finetuning module, the network parameters of the pretrained graph model are frozen, and a groups of learnable prompt vectors assigned to each graph-level representation are trained for adapting different target graph-level datasets with various data distributions. Empirical experiments conducted on several benchmark graph-level datasets further indicate the effectiveness and generalization capability of the proposed GPC method.

Graph-level clustering is a recently emerging subfield of research, due to the inherent complexity and high variability of the molecular datasets, particularly challenging clustering is made according to the comparison results, with low NMI and ARI values on some molecular datasets. This further demonstrates that the current mechanism design may struggle to capture the complex structural and semantic patterns inherent in molecular graphs. In the future work, we will further investigate the characteristics of the graph-level datasets, and attempt to incorporate some domain-specific knowledge to improve the discriminativeness of graph-level representations and the clustering outcomes.

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